Basis and dimentions

Friday, June 19, 2026 10:14 AM

R^2

S is basis

① vector Should be linearly Independent

② "S" Should Span R^2

$$V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad V_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$$K_1 V_1 + K_2 V_2 = 0$$

$$K_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + K_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} = 0$$

$$\begin{pmatrix} K_1 + 0K_2 \\ 0K_1 + K_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$K_1 = 0, \quad K_2 = 0$

Independent

$$\begin{pmatrix} a \\ b \end{pmatrix} = K_1 V_1 + K_2 V_2$$

$$\begin{pmatrix} a \\ b \end{pmatrix} = \underline{K_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + K_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix}} \Rightarrow \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} K_1 + 0K_2 \\ 0K_1 + 1K_2 \end{pmatrix}$$

$a = K_1$

$b = K_2$

$$\begin{pmatrix} a \\ b \end{pmatrix} = a V_1 + b V_2$$

Example 3: Show that the vectors $v_1 = (1,2,1), v_2 = (2,9,0), v_3 = (3,3,4)$ form basis for R^3.

① Linear Independance

$$K_1 V_1 + K_2 V_2 + K_3 V_3 = 0$$

$$K_1 \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} + K_2 \begin{pmatrix} 2 \\ 9 \\ 0 \end{pmatrix} + K_3 \begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix} = 0$$

$$K_1 + 2K_2 + 3K_3 = 0$$
$$2K_1 + 9K_2 + 3K_3 = 0$$
$$K_1 + 0K_2 + 4K_3 = 0$$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 2 & 9 & 3 & 0 \\ 1 & 0 & 4 & 0 \end{array}\right)$$

$$\begin{array}{l} R_2 - 2R_1 \\ R_3 - R_1 \end{array} \left(\begin{array}{ccc|c} 1 & 2 & 3 & 0 \\ 0 & 5 & -3 & 0 \\ 0 & -2 & 1 & 0 \end{array}\right)$$

$$5K_2 - 3K_3 = 0$$
$$-2K_2 + K_3 = 0$$

$$R_2 + 2R_3 \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & -2 & 1 & 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 & 1 & \\ 0 & -2 & 1 & 0 \end{pmatrix}$$

$$R_3+2R_1 \begin{array}{c} \\ \begin{pmatrix} 1 & 2 & 3 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 3 & 0 \end{pmatrix} \end{array}$$

K_3

$3K_3 = 0$

$\boxed{K_3 = 0}$

$K_2 + K_3 = 0$

$K_2 + 0 = 0$ $\boxed{K_2 = 0}$

$K_1 + 2K_2 + 3K_3 = 0$

$K_1 = 0$

$K_1 = K_2 = K_3 = 0$

Linearly Independent

Spanning

$$K_1 + 2K_2 + 3K_3 = a$$
$$2K_1 + 9K_2 + 3K_3 = b$$
$$K_1 + 0K_2 + 4K_3 = c$$

$$\begin{vmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{vmatrix}$$

$$= 1\begin{vmatrix} 9 & 3 \\ 0 & 4 \end{vmatrix} - 2\begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} + 3\begin{vmatrix} 2 & 9 \\ 1 & 0 \end{vmatrix}$$

$= 36 - 2(5) + 3(-9)$

$= 36 - 10 - 27$

$= -1 \neq 0$

$$\left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 2 & 9 & 3 & b \\ 1 & 0 & 4 & c \end{array}\right)$$

$$\begin{array}{l} \\ R_2-2R_1 \\ R_3-R_1 \end{array} \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 5 & -3 & \boxed{b-2a} \\ 0 & -2 & 1 & c-a \end{array}\right)$$

$b - 2a + 2c - 2a$

$$R_2+2R_3 \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & -1 & b+2c-4a \\ 0 & -2 & 1 & c-a \end{array}\right)$$

$1 - 2$

$c - a - 8a + 2b + 4c$

$$R_3+2R_2 \left(\begin{array}{ccc|c} 1 & 2 & 3 & a \\ 0 & 1 & \boxed{-1} & -4a+b+2c \\ 0 & 0 & +1 & +9a \mp 2b \mp 5c \end{array}\right)$$

$K_3 = 4a - 2b - 5c$

$K_2 = -4a + b + 2c + K_3$

Example 4: **Let** $\boldsymbol{v_1 = (1,1), v_2 = (3,5), v_3 = (4,2)}$.**Check** whether v_1, v_2, v_3 form basis for R^2 or not?

$$k_1 v_1 + k_2 v_2 + k_3 v_3 = \vec{0}$$

$$k_1(t^2+1) + k_2(t-1) + k_3(2t+2) = 0t^2 + 0t + 0$$

$$K_1 t^2 + K_1 + \underline{K_2 t} - K_2 + \underline{2K_3 t} + 2K_3 = 0t^2 + 0t + 0$$

$$K_1 t^2 + (K_2 + 2K_3)t + (K_1 - K_2 + 2K_3) = 0t^2 + 0t + 0$$

$$K_1 = 0$$

$$K_2 + 2K_3 = 0$$

$$K_1 - K_2 + 2K_3 = 0$$

① If $A = \begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix}$; $I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

find "λ" if $\det(A - \lambda I) = 0$ ✓

$$\begin{pmatrix} 4 & 2 \\ 3 & -1 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix}$$

$$= \begin{pmatrix} 4-\lambda & 2 \\ 3 & -1-\lambda \end{pmatrix}$$

$$\begin{vmatrix} 4-\lambda & 2 \\ 3 & -(\lambda+1) \end{vmatrix} = 0$$

$-(\lambda+1)(4-\lambda) - 6 = 0$

$(\lambda+1)(4-\lambda) + 6 = 0$

$-\lambda^2 + 4\lambda - \lambda + 4 + 6 = 0$

$-\lambda^2 + 3\lambda + 10 = 0$

Charateristic equation $\boxed{\lambda^2 - 3\lambda - 10 = 0}$

Eigen values $\boxed{\lambda = -2, 5}$

0 3 5 9

$$\begin{pmatrix} 0 & 3 \\ 5 & 9 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 5 \\ 3 & 9 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} -\lambda & 5 \\ 0 & 9-\lambda \end{pmatrix}$$

$$\begin{pmatrix} -\lambda & 3 \\ 5 & 9-\lambda \end{pmatrix}$$

$-\lambda(9-\lambda) - 0 = 0$

$-\lambda(9-\lambda) = 0$

$\lambda = 0,$

$-\lambda(9-\lambda) - 15 = 0$

$\lambda(9-\lambda) + 15 = 0$

$9\lambda - \lambda^2 + 15 = 0$

$-\lambda^2 + 9\lambda + 15 = 0$

$\lambda^2 - 9\lambda - 15 = 0$

$b^2 - 4ac \geq 0$

$\frac{9 \pm \sqrt{141}}{2}$

$(-9)^2 - 4(1)(-15)$

$\frac{9+\sqrt{141}}{2}, \quad \frac{9-\sqrt{141}}{2} = 81 + 60$

(-4)

$$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0$$

$\lambda = 4;$

$$
\begin{array}{c|cccc}
4) & 1 & -8 & 17 & -4 \\
 & \downarrow & 4 & -16 & 4 \\
\hline
 & 1 & -4 & 1 & 0
\end{array}
$$

-4

$\pm 1, \pm 2, \pm 4$

$\pm \frac{1}{2}, \pm \frac{1}{4}$

$\lambda^2 - 4\lambda + 1 = 0$

$= + \frac{4 \pm \sqrt{16 - 4}}{2} = \frac{4 \pm \sqrt{12}}{2} = \frac{4 \pm 2\sqrt{3}}{2}$

$= 2 + \sqrt{3},\ 2 - \sqrt{3}$

$\lambda^3 - 8\lambda^2 + 17\lambda - 4 = 0$

Example 4: Find the eigenvalues and the corresponding eigenvectors of the following matrix.

$A = \begin{bmatrix} 3 & 0 \\ 8 & -1 \end{bmatrix}$

Solution: The eigenvalue/s of A are the solution of the equation

$det(A - \lambda I) = 0$

$det(A - \lambda I) = \begin{vmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{vmatrix}$

$-(3-\lambda)(1+\lambda) = 0$

$\lambda = 3, \quad \lambda = -1$

If for $\lambda = 3$, eigenvector is $\begin{bmatrix} x \\ y \end{bmatrix}$, then

$A\vec{x} = \lambda\vec{x}$

$(A - \lambda I)\vec{x} = \vec{0}$

Using $\lambda = 3$ in

$\begin{bmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$\begin{bmatrix} 0 & 0 \\ 8 & -4 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$0x + 0y = 0$

$8x - 4y = 0$

$8x = 4y \Rightarrow x = \frac{1}{2}y$

Let $y = t$,

$x = \frac{1}{2}t$

$A\vec{x} = \lambda\vec{x}$

$(A - \lambda I)\vec{x} = 0$

$\begin{pmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$

$\begin{pmatrix} 0 & 0 \\ 8 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$

$0x + 0y = 0$

Let $y = t$,

$x = \frac{1}{2}t$

$\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} \frac{1}{2}t \\ t \end{bmatrix} = t\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$

So, $\begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = 3$.

$0x + 0y = 0$

$8x - 4y = 0$

$x = \frac{1}{2}y$

$y = t$

$x = \frac{1}{2}t$

$\begin{pmatrix} \frac{1}{2}t \\ t \end{pmatrix} = t\begin{pmatrix} \frac{1}{2} \\ 1 \end{pmatrix}$

Now using $\lambda = -1$,

$\begin{bmatrix} 4 & 0 \\ 8 & 0 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$4x + 0y = 0 \quad \dots (1)$

$8x - 0y = 0 \quad \dots (2)$

$\Rightarrow \; x = 0$ *from both equations* (1) and (2)

$\begin{pmatrix} 3-(-1) & 0 \\ 8 & -1-(-1) \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{matrix} 0 \\ 0 \end{matrix}$

As y is free variable so put $y = t$, and $\begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ t \end{bmatrix} = t\begin{bmatrix} 0 \\ 1 \end{bmatrix}$

So, $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ is the eigenvector corresponding to $\lambda = -1$.

$\begin{vmatrix} 3-\lambda & 0 \\ 8 & -1-\lambda \end{vmatrix}$

$\left.\begin{matrix} 4x = 0 \\ 8x = 0 \end{matrix}\right\} x = 0$

$\begin{pmatrix} 4 & 0 \\ 8 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\begin{pmatrix} 0 \\ t \end{pmatrix} = t\underbrace{\begin{pmatrix} 0 \\ 1 \end{pmatrix}}$ eigen vector

Linear Combination
depend
span

~~Basis~~
eigen value
eigen vec